Ans : (b) Mammals are a class of animals, which are differentiated from other classes of animals such a reptile, birds, fish, etc. by the presence of hair, three middle ear bones, mammary glands in females, among others. While Sea cow (also known as Dugong) and Sea lion are examples of mammals, Sea horse is a vertebrate fish.

• Agriculture, Animal Husbandry, Dairy

79. Fruits stored in a cold chamber exhibit longer storage life because –
(a) exposure to sunlight is prevented
(b) concentration of carbon dioxide in the environment is increased
(c) rate of respiration is decreased
(d) there is an increase in humidity

Ans : (c) The stored fruits in the cold cell last longer because the rate of respiration decreases as temperature drops, which reduces the decaying capacity and therefore the, food in the refrigerator does not spoil quickly.

80. Consider the following crops :
1. Cotton 2. Groundnut
3. Rice 4. Wheat
Which of these are Kharif crops ?
(a) 1 and 4 (b) 2 and 3 only
(c) 1, 2 and 3 (d) 2, 3 and 4

Ans : (c) In India, the cultivation seasons are divided into Kharif and Rabi. Wheat is a Rabi crop since it requires a lower temperature. All the other three are Kharif crops. Kharif Crops: Paddy (Rice) Maize, Groundnut, Sugarcane, Horse grain, Jute, Flax, Cotton etc.
Rabi Crops: -Wheat , Barley, Oats, tauria (lahi) and mustard, Sun Flower etc.

81. Many transplanted seedlings do not grow because
(a) the new soil does not contain favourable minerals
(b) most of the root hairs grip the new soil too hard
(c) most of the root hairs are lost during transplantation
(d) leaves get damaged during transplantation

Ans : (c) Newly transplanted trees or shrubs experience some degree of transplant shock. This is simply the plant reacting to having been moved. It is generally caused by damage to the roots of the plants during transplant. While the thickest roots are nearest to the root ball, the most important roots, those that actually "do the work", are located the farthest away from the plant. These tiny roots are covered by even tinier hairs which absorb most of the water that is eventually carried to the top growth of the plant.

82. Improper handling and storage of cereal grains and oilseeds result in the production of toxins known as aflatoxins which are not generally destroyed by normal cooking process. Aflatoxins are produced by
(a) bacteria (b) Protozoa
(c) moulds (d) viruses

Ans: (c) Aflatoxins are naturally occurring mycotoxins, which are produced by many species of Aspergillus, a fungus. The fungus can be considered a type of mould. Aflatoxins are toxic and are among the most carcinogenic substances known.

83. Which of the following adds/ add nitrogen to the soil ?
1. Excretion of urea by animals
2. Burning of coal by man
3. Death of vegetation
Select the correct answer using the codes given below :
(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans : (c) Waste products of animals, as well as dead remains of organism, are converted into inorganic ammonia by bacteria in the process of ammonification. This ammonia is then converted into nitrate then nitrites by Nitrosomonas and Nitrococcus bacteria, respectively. Another soil bacteria Nitrobacter can convert nitrate into nitrite. In this way, the process of nitrogen cycle continues and the amount of the nitrogen in the soil is replenished. Statement 2 is wrong as the burning coal does not produce nitrogen.

• Science & Technology

84. Mycorrhizal biotechnology has been used in rehabilitating degraded sites because mycorrhiza enables the plants to –
1. resist drought and increase absorptive area
2. tolerate extremes of pH
3. resist disease infestation
Select the correct answer using the codes given below :
(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans : (d) Mycorrhizal biotechnology is used in the rehabilitation of degraded sites. This gives plants the ability to resists drought and increases absorptive area. Also, the ability to tolerate pH extremes and resistance to disease increases.

85. Consider the following organisms :
1. Agaricus
2. Nostoc
3. Spirogyra
Which of the above is /are used as bio fertilizer/bio –fertilizers ?.
(a) 1 and 2 (b) 2 only
(c) 2 and 3 (d) 3 only

Ans : (b) Bioinoculants are the microbial preparation that enhances the uptake of nutrients by plants from the soil, increase the availability of the nutrient and stimulate plant growth.
There are two types of organisms that are used as inoculants:
(i) Symbolic organisms (such as Rhizobium, Synorhizobium, Bradyrhizobium, Nostoc, Anabaena etc..) and
(ii) Non-Symbiotic organism (such as Azoto-bacter, Azospirillum, Beijerinckia, etc.) Agaricus is a genus of mushrooms, both edible and poisonous species.
Spirogyra is a genus of filamentous algae of the order of zygnematales, named for the helical or spiral arrangement of the chloroplasts that is diagnostic of the genus.

ENVIRONMENT & ECOLOGY

86. With reference to the food chains in ecosystems. Which of the following kinds. of organism is/ are known as decomposer organism/organisms ?
1. Virus 2. Fungi 3. Bacteria
Select the correct answer using the codes given below :
(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans : (b) Viruses are not decomposers. They are like parasites and live off on a living organism. Fungi and Bacteria, on the other hand, are decomposers and perform the important role of breaking down the dead remains of plants and animals into nutrients.

87. In the grasslands, trees do not replace the grasses as a part of an ecological succession because of
(a) insects and fungi
(b) limited sunlight and paucity of nutrients
(c) water limits and fire
(d) None of the above

Ans : (c) In a grassland, trees have already been cleared away due to fires in the past, which also changes the ground composition. Without the trees to hold them, water washes away most of the soil and the underlying ground becomes rocky and barren. After this, lichens become the pioneer population, which is then replaced by grasses. The grasses, however absorb most of the water before it reaches the roots of the shrubs. Thus, the ecological succession does not occur beyond the grasses.

88. Consider the following fauna of India :
1. Gharial 2. Leatherback turtle
3. Swamp deer
Which of the above is/are endangered ?
(a) 1 and 2 only (b) 3 only
(c) 1, 2 and 3 (d) None

Ans : (c) In Indian animals, Gharial have been declared as endangered species by the IUCN. The world's most endangered Sea turtle is a leatherback turtle found near Nicobar Island and Swamp deer found in Central and Northern India is also an endangered species. Hence the correct answer will be (c).

89. Due to improper/ indiscriminate disposal of old and used computers or their parts, which of the following are released into the environment as e-waste ?
1. Beryllium 2. Cadmium
3. Chromium 4. Heptachlor
5. Mercury 6. Lead
7. Plutonium
Select the correct answer using the codes given below
(a) 1, 3, 4, 6 and 7 only
(b) 1, 2, 3, 5 and 6 only
(c) 2, 4, 5 and 7 only
(d) 1, 2, 3, 4, 5, 6 and 7

Ans : (b) Beryllium is used in Circuits board, Hard disk drive, Motherboard, wiring board and connectors. So option (c) is wrong.
Mercury is a key part of some modern technologies, including LCD Screens, Laptops and Computers, so option (A) is also wrong.
Heptachlor is an organochlorine compound that was used as an insecticide. So option (d) is also wrong. So, the correct answer is (b).

90. Acid rain is caused by the pollution of environment by
(a) carbon dioxide and nitrogen
(b) carbon monoxide and carbon dioxide
(c) ozone and carbon dioxide
(d) nitrous oxide and sulphur dioxide

Ans : (d) Nitrous acid and sulphur dioxide are mainly responsible for acid rain. Nitrogen and sulphur oxides react with water to make ammonium nitrate and sulfuric acid, respectively and they cause damage in the form of rain.

91. With reference to food chain in ecosystems, consider the following statements :
1. A food chain illustrates the order in which a chain of organisms feed upon each other.
2. Food chain are found within the populations of a species.
3. A food chain illustrates the numbers of each organism which are eaten by others.
Which of the statements given above is/are correct?
(a) 1 only (b) 1 and 2 only
(c) 1, 2 and 3 (d) None

Ans : (a) A food chain is a sequence of who eats whom in a biological community (an ecosystem) to obtain nutrition.

92. Consider the following pairs :

National Park		River flowing through the Park
1. Corbett National Park	–	Ganga
2. Kaziranga National Park	–	Manas
3. Silent Valley National Park	–	Kaveri

Which of the above pairs is/are correctly matched ?
(a) 1 and 2 (b) 3 only
(c) 1 and 3 (d) None

Ans : (d) Ramganga, a tributary of Ganga, passes through Corbett National Park. While silent valley National Park is located in the state of Kerala, the Kunthipuzha river flows through this park and the Brahmaputra flows through Kaziranga National Park (Assam).

93. In which of the following States is lion-tailed macaque found in its natural habitat ?
1. Tamil Nadu 2. Kerala
3. Karnataka 4. Andhra Pradesh
Select the correct answer using the codes given below :
(a) 1, 2 and 3 (b) 2 only
(c) 1, 3 and 4 (d) 1, 2, 3 and 4